

TagSort: Accurate Relative Localization Exploring RFID Phase Spectrum Matching for Internet of Things

Jinjiang Lai, Chengwen Luo[✉], *Member, IEEE*, Jiawei Wu, Jianqiang Li[✉], Jia Wang[✉], Jie Chen, Gang Feng, and Houbing Song[✉], *Senior Member, IEEE*

Abstract—The radio frequency identification (RFID) technologies, which have been widely adopted in different Internet of Things (IoT) applications, are the fundamental building block for achieving smart factories, smart logistics, smart stores, etc. Besides knowing the ID of tags, the relative location information of different tags is of great importance since it contains the spatial relationship among different tags, which is essential for many object localization applications beyond the capacity of absolute localization approaches. In this article, we propose *TagSort*, an RFID-based sorting system that exploits the physical layer information, i.e., the phase of RFID wireless signals to achieve the relative localization of different tags. Several novel filtering and peak detection algorithms are proposed to achieve accurate and robust detection of the order of tags. Extensive evaluation shows promising results (over 95% accuracy) and make *TagSort* a promising system for future RFID sorting systems, thus enabling a variety of IoT applications and services.

Index Terms—Indoor localization, phase spectrum, relative localization, radio frequency identification (RFID).

I. INTRODUCTION

OVER the past decade, radio frequency identification (RFID) technology has been widely adopted in a variety of applications, such as asset management, access control, object localization, and tracking, etc. [1], [2]. Among these applications, the RFID-based positioning systems have attracted extensive attention mainly because they can be easily deployed at a large scale with low cost. It is envisioned that the RFID-based positioning systems will continue to gain popularity with the rise of the Internet of Things (IoT) [3]–[5].

There are a large body of work on RFID-based indoor localization system in [3] and [6]–[11]. Based on the technologies they used, existing works can be classified into

three groups: 1) received signal strength (RSS)-based methods [10], [11]; 2) arrival of angel (AOA)-based methods [12], [13]; and 3) phase-based methods [8]. Despite these efforts, prior work mainly focused on absolute localization, and there is limited work investigating RFID-based relative localization. However, many object localization applications need the relative locations for a set of objects instead of their absolute locations [14]. For example, when looking for misplaced books in a library, we need to get the current relative order of books on the bookshelf instead of knowing their absolute coordinates. The absolute localization approaches cannot be directly applied in the relative localization system for two reasons. First, most absolute object schemes [3], [6]–[9] use RF signal information for localization, such as signal propagation speed, signal intensity, and signal propagation direction. Due to the complex indoor environment and significant multipath effect, the localization accuracy is hard to remain high. For example, the major advantage of LANDMARC [10] is that it improves the overall accuracy of locating objects by utilizing the concept of reference tags. However, the error due to the dynamics of the environment can hardly be alleviated. Second, in order to achieve high localization accuracy, one has to use high cost hardware and software support (e.g., USRP), or depend on the scene (e.g., location fingerprint-based techniques [15]).

Recently, Shangguan *et al.* [14] proposed the first RFID-based relative localization system by utilizing the phase profile. The idea is simple but effective: when we move the RFID reader along one direction, the distance between the reader and the tag will first decrease and then increase, and achieve the minimum when the reader is perpendicular above the tag, so is the phase profile. Based on this observation, they design an RFID-based relative localization using spatial-temporal phase profiling. This method, though efficient, still encounters a number of problems in real environments. For example, the feature zone detection method in [14] is susceptible to self-interference and environmental noise. To encounter this problem, in this article we propose *TagSort*, a high-accuracy relative localization system exploiting the RFID phase spectrum. Different from the current state of the art, *TagSort* proposes a series of novel signal processing methods to improve the robustness of feature zone detection, so as to improve the overall robustness of the RFID-based relative localization system. Besides, *TagSort* further incorporates a dynamic template selection algorithm to address the distance

Manuscript received April 26, 2019; revised September 15, 2019; accepted October 14, 2019. Date of publication October 30, 2019; date of current version January 10, 2020. This work was supported in part by the National Natural Science Foundation of China under Grant 61972263, Grant 61602319, and Grant U1713212; and in part by the Natural Science Foundation of Guangdong Province under Grant 2017A030313357. (Corresponding author: Chengwen Luo.)

J. Lai, C. Luo, J. Wu, J. Li, J. Wang, J. Chen, and G. Feng are with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518000, China (e-mail: chengwen@szu.edu.cn)

H. Song is with the Department of Electrical, Computer, Software, and Systems Engineering, Embry-Riddle Aeronautical University, Daytona Beach, FL 32114 USA.

Digital Object Identifier 10.1109/IIOT.2019.2950174

variations during the tag sorting. We implement and evaluate the system, which shows promising results. Our contributions are summarized as follows.

- 1) We propose a hybrid feature zone detection and peak detection algorithm to robustly detect the feature zones and peaks which appear in the RF phase profile. Different from the state-of-the-art [14], *TagSort* utilizes the whole phase spectrum instead of using the single peaks detected, which efficiently improves the robustness of the relative localization.
- 2) We incorporate a dynamic template selection algorithm to encounter the distance variations between readers and the tags, which addresses the current drawbacks in the state of the art, which assumes the constant distances during the relative localization process.
- 3) We implement and evaluate the system in dynamic real-world environments, which shows promising results with 95% sorting accuracy even the tags are closely placed. The results show that the proposed system has the potential to be used in future practical tag sorting systems.

The rest of this article is organized as follows. First, Section II discusses the related work. Section III introduces technical background, and Section IV describes our system. Followed by that, Section V discusses evaluation results, and Section VI discusses limitation and future work. Finally, Section VII concludes this article.

II. RELATED WORKS

Indoor localization is an active research area due to its numerous application scenarios and has attracted many research works in [16]–[18]. Among these works, most research of indoor localization technology studies the indoor absolute localization technologies, with the goal of finding the accurate object location in the physical coordinate system. Many different interfaces are used, including WiFi, Bluetooth, ZigBee, UWB, ultrasound, etc. RFID, as one of the fundamental enablers of the IoT, has been widely used in many IoT systems, and the signal of which has also been exploited in the localization systems as well [10], [14]. Due to its pervasive usage and battery-free property, RFID-based localization systems continue to gain popularity in both academia and industry. This article focuses on the RFID-based localization techniques and is closely related to several categories of work in the literature that exploits RFID signals for localization. In this section, we mainly discuss the recent developments in the related *RSSI-based* localization approaches and *phase-based* localization approaches.

A. RSSI-Based Approaches

The RSSI-based localization approaches adopt the similar idea of WiFi-fingerprinting [19], [20], and the fundamental idea of which is that wireless signal strengths decrease as the signal propagation distance increases, therefore the RSSI signals of the RFID tags can be used as range indicators to determine the location of the tags. Most of the research works in this category require typically predeployment of

many reference RFID tags around the area-of-interest, and then use the RSSI values of these reference tags to locate a specific tag [10], [11]. However, the major limitation of RSSI-based methods is that they are usually high sensitive to multipath effects, and thus difficult to achieve high-precision localization in dynamic environments. OTrack [7] is a localization scheme for moving objects in a fixed orbit environment, which tracks the baggages in the airport by processing the time-varying RSSI information of tags attached to the baggages. But such schemes often require specific working scenarios and their accuracy can be easily affected once the environments are changed. PinIt [11] proposes a fine-grained RFID localization scheme in a multipath and nonline-of-sight scenarios, but the method is mainly used for locating the static RFID tags, and the performance can degrade in tracking the moving tags.

B. Phase-Based Approaches

In recent years, researchers have started to investigate the phase-based localization approaches since phase measurements are usually of higher resolution than RSSI and are sensitive to location changes, making it possible to achieve high localization accuracy in dynamic environments. Recently phase-based localization approaches have attracted the attention of many researchers. MobiTagbot [8] uses antenna movements and frequency hopping to simulate a set of virtual antennas and construct a holography to locate RFID tags. However, as the monitoring area becomes bigger, the computation increases and has a significant impact on the real-time localization performance. STPP [14] first proposes the concept of *relative localization* and tries to find out the accurate tag relative ordering by analyzing the characteristics of the symmetric change of the phase information of RFID tags when the reader approaches and moves away. This article is appealing since it opens up a new way for simple but effective for tag ordering. However, several drawbacks also exist.

- 1) The relative localization approach proposed in [14] highly relies on the peaks detected in the phase profiles, making it susceptible to environmental noises caused by the multipath effect and environmental dynamics such as human movements.
- 2) The localization approach proposed in [14] assumes the constant distance between the reader and the tags, when the distance changes in the localization process, the performances can easily get degraded.

This article is closely related to the above phase-based localization approaches with the aim of finding the accurate ordering of tags using the relative localization techniques. To address the current limitations in the state-of-the-art, we propose the *TagSort* system to incorporate a hybrid approach for feature zone detection and peak detection. To improve the robustness and accuracy, *TagSort* utilizes the whole spectrum information for correlation matching and uses a dynamic template selection to address the distance variations between the reader and the tags. Different from the current state of the art, the *TagSort* system automatically adapts to the distance changing between the reader and the tags, and provides high relative localization accuracy even in dynamic environments.

III. TECHNICAL BACKGROUND

In this section, we briefly introduce the technical background of ultrahigh-frequency (UHF) RFID systems and their signal propagation information that affects the design of indoor relative localization. We are particularly interested in phase information, which is the fundamental fine-grained information that is used for *TagSort* relative localization. Before introducing *TagSort*, for the sake of convenience we summarize the notations in Table I which will be adopted in the later algorithm design.

A. RFID Working Principles

The RFID system communicates via electromagnetic waves, and the starting point for all of its work comes from the RF signals generated by the reader. The reader sends an RF signal of a specific frequency to the space through the transmitting antenna to form an electromagnetic field and waits for feedback. The electronic tag in the electromagnetic field receives the RF signal to generate an induced current, so that the obtained energy is activated. The signal containing the self-encoded information is loaded into the carrier signal and transmitted to the space through the antenna of the tag, aiming to feedback information to the antenna through the echo wave. This process will continue as long as the tag is in the electromagnetic field of the reader's transmitting antenna and sufficient energy is available. The receiving antenna of the reader receives the echo modulated signal sent from the tag and transmits it to the reader signal processing module via the antenna adjuster. After demodulation and decoding, the valid information of the tag is obtained and sent to the host system. The host system recognizes the identity of the tag according to the logical operation and performs corresponding processing and control according to different settings and applications, and finally issues a command signal to control the reader to complete the corresponding operation.

B. Key Localization Related RFID Wireless Signal Properties

Phase is the basic property of wireless signals. The phase value of an RF signal describes the degree of deviation between the receiving signal and the sending signal, ranging from 0 to 2π , which is determined by the transmitting circuit of the reader, the spatial distance, the reflection characteristics of the RF tag, and the receiving circuit of the reader [9]. Compared with RSSI, RFID phase information is demonstrated to be more robust to environmental interference [9], [21]. We know that for a carrier with a frequency f (Hz), the wavelength λ and frequency f , it satisfies the following relationship:

$$\lambda = \frac{c}{f} \quad (1)$$

where c is the propagation velocity of electromagnetic waves in the propagation medium, in the air $c \approx 3.0 \times 10^8$ m/s. The signal transmitted from the reader is returned to the reader through the reflection of the tag, during which a total of $2R$ (R is the spatial distance between the reader and the tag), coupled with the phase shift caused by the reader's transmit circuit,

TABLE I
SUMMARY OF NOTATIONS

Notations	Meaning
λ	the wave length
c	the propagation velocity of electromagnetic waves in the propagation medium, in the air $c \approx 3.0 \times 10^8$ m/s
f	the frequency
θ	the theoretical phase
θ_T	the phase shift caused by reader's transmit circuit
θ_R	the phase shift caused by reflection characteristics of tag
θ_{TAG}	the phase shift caused by reader's receiving circuit
R	the spatial distance between the reader and tag
d	the distance between the tag and the reader
L_0	the vertical distances between the reader and the tag
D_0	the horizontal distances between the reader and the tag
V	the moving velocity
t	the moving time
θ_t	the RF phase
θ_x	the sample phase that requires to be filtered
T_1, T_2, T_3, T_4	the average of the timestamp of these four sets of four methods
N_i	the size of that set
$t_{i,k}$	the time stamp of k -th data point in the i -th set
R_i	contains N_i samples
$\bar{T}_i$	the average timestamp in region R_i
ω_1	the goodness of matching between the theoretical template and reference points
ω_2	the uniformity of moving speed

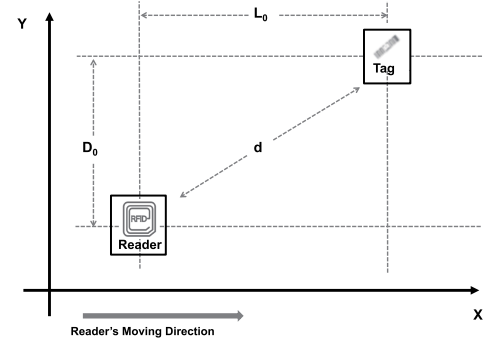


Fig. 1. Illustration of detection process.

reflection characteristics of tag, and reader's receiving circuit. Without considering other interference factors, the theoretical RF phase can be determined by

$$\theta = 2\pi \left(\frac{2R}{\lambda} \right) + \theta_T + \theta_R + \theta_{TAG} \quad (2)$$

where θ_T , θ_R , and θ_{TAG} represent the phase shift caused by reader's transmit circuit, reflection characteristics of tag and reader's receiving circuit, respectively. Considering that the phase value θ is a periodic function with a period of 2π , (2) can be written as follows [14]:

$$\begin{cases} \theta = (2\pi \left(\frac{2R}{\lambda} \right) + \mu) \mod (2\pi) \\ \mu = \theta_T + \theta_R + \theta_{TAG}. \end{cases} \quad (3)$$

C. Intuition Behind the Proposed Scheme

The basic idea of our approach is based on the observation in [14]: Given a static RFID tag, as we move the reader over the tag along one direction, the larger the distance between the reader and the tag, the smaller the phase changes as the reader moves. For example, as shown in Fig. 1,

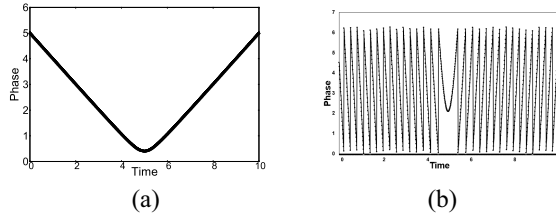


Fig. 2. Phase profile of a single tag. (a) Distance between the reader and a single tag. (b) Theoretical time series of RF phase over a single tag.

suppose there is a reader and a tag. The tag is stationary and we move the reader along the direction as shown in Fig. 1. The distance between the reader and a tag varies in the following way (which can be easily known from Fig. 1):

$$d = [(L_0 - Vt)^2 + (D_0)^2]^{\frac{1}{2}} \quad (4)$$

where d is the distance between the tag and the reader at time t , and L_0 and D_0 are the vertical and horizontal distances between the reader and the tag. After taking (3) into consideration, the pattern in which the RF phase changes with time is theoretically as follows:

$$\theta_t = \left\{ \frac{4\pi}{\lambda} [(L_0 - Vt)^2 + (D_0)^2]^{\frac{1}{2}} + \mu \right\} \bmod (2\pi). \quad (5)$$

If we assume $\mu = 0$ rad, $V = 1$ m/s, $L_0 = 5$ m, $D_0 = 0.20$ m, and $\lambda = 0.30$ m, RF phase θ_t will theoretically change as shown in Fig. 2(a) and (b). According to (5), the phase of the received signal will first decrease then increase.

Since the range of any phase value is $[0, 2\pi)$, when the phase value decreases to 0, it will immediately jump to 2π and start decreasing from 2π . Similarly, when the phase value increases to 2π , it will immediately drop to 0 and then increase from that again. This phenomenon holds when the reader moves from one side to another side except when the reader is perpendicular above the tag. When the reader is close to the tag, it will generate a U shape which spans the whole range of $[0, 2\pi)$. The bottom of U shape is the time when the reader is right above the tag [14]. Based on this observation, given a sequence of tags, we can obtain their relative location by moving a reader from the beginning to the end along with their array direction. For example, if tag 1 appears before tag 2 then the phase bottom of tag 1 should be detected earlier before the bottom of tag 2. However, in reality, the phase profile contains much noise because of multipath self interference [22] and environmental noise, it is difficult to detect the bottom point precisely. In this article, we propose a scheme to address this challenge, and the details are provided in the next section.

IV. SYSTEM DESIGN

To determine the accurate ordering of tags, our goal is to accurately detect and find the order of bottom points of U shape in phase profiles. To achieve that, data preprocessing is required to remove the noises contained in the phase profiles. As shown in Fig. 3, the proposed system includes the following four stages: 1) data filtering; 2) U zone detection; 3) bottom

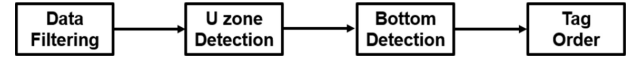


Fig. 3. System flowchart.

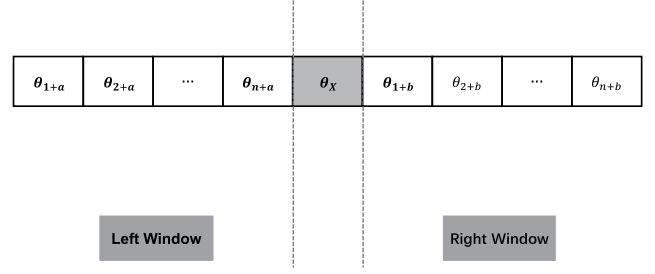


Fig. 4. Double windows moving average filter.

detection; and 4) tag order determination. In the following parts, we will describe the design details of each component.

A. Data Filtering

As mentioned earlier, the raw phase profile contains much noise that will deteriorate the bottom detection accuracy. A traditional method is applied for moving average filter to remove noise. However, the traditional moving average filter works as follows. A window with n data points is maintained, and every time a new data point arrives, the first data point in the window will be removed and the new data point will be added to the end of the window, and the filtering result of this turn will be the average value of all data point in the window. We know the theoretical time series without noisy data are bilaterally symmetrical, and there is a quasi-periodic characteristic implied on each side of the symmetrical center (the so-called bottom point). We also know there are "jump" points, such as 0 to 2π jumps and 2π to 0 jumps legally existing in the time series, which causes the original moving average filter to fail.

To solve this problem, we design a modified moving average filter. Compared to the Kalman filter, this algorithm is lightweight and can effectively remove noise points caused by environmental noise and multipath effect, and at the same time it will not be affected by the legitimate "jumping" points in the phase profiles. The steps are illustrated in Fig. 4, θ_x represents the sample that requires to be filtered, and we maintain two windows with n data points on the left side and right side of θ_x . Before filtering, we need to first check if θ_x satisfies the following restrictions:

$$|\theta_x - \frac{\sum_{i=1}^n \theta_{a+i}}{n}| \leq \tau \quad (6)$$

$$|\theta_x - \frac{\sum_{i=1}^n \theta_{b+i}}{n}| \leq \tau \quad (7)$$

where τ is the threshold, and $((\sum_{i=1}^n \theta_{a+i})/n)$ and $((\sum_{i=1}^n \theta_{b+i})/n)$ find the mean of the left and right windows, respectively. If the mean is less than the threshold, we will perform normal moving averaging filter on θ_x , and move left/right window to the next sample. Otherwise, θ_x is regarded as noise and will be removed from the data.

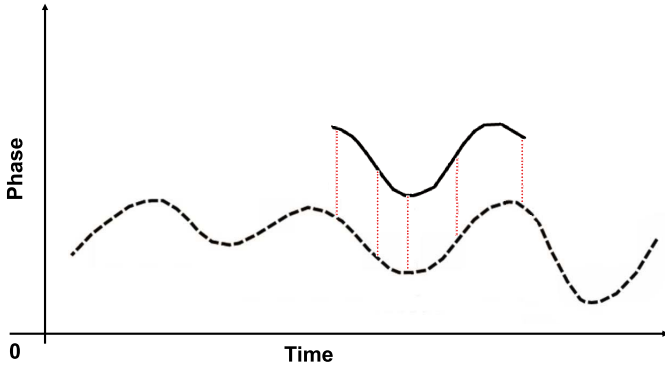
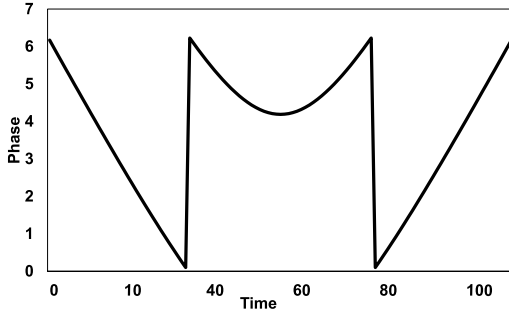
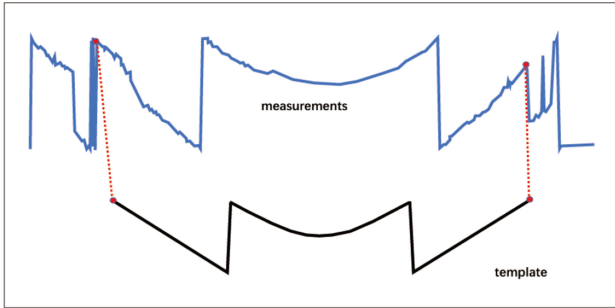


Fig. 5. Illustration of subsequence DTW.

Fig. 6. Template for $\cup$ zone matching.Fig. 7. Illustration of $\cup$ zone detection using subsequence DTW.

B. $\cup$ Zone Detection

After data filtering, we need to detect $\cup$ zone which denotes the time period during which the reader is right above the tag. Previous work [14] uses dynamic time warping (DTW) algorithm [23], [24]. However, we find that the original DTW algorithm cannot be directly applied in our scheme because here it needs to detect $\cup$ zone from the entire time series. To this end, applying subsequence DTW [25] becomes a better option. Subsequence DTW is designed to find a subsequence within a longer given sequence which fits another sequence (query sequence) best. Fig. 5 is a typical example of how subsequence DTW works.

To apply subsequence DTW, we first build a template using the theoretical expression of the featured zone as illustrated by (5) as shown in Fig. 6. Fig. 7 shows the result of $\cup$ zone detection using subsequence DTW. We can see that the measured phase profile can match with the theoretical template very well, and the $\cup$ zone can be accurately detected.

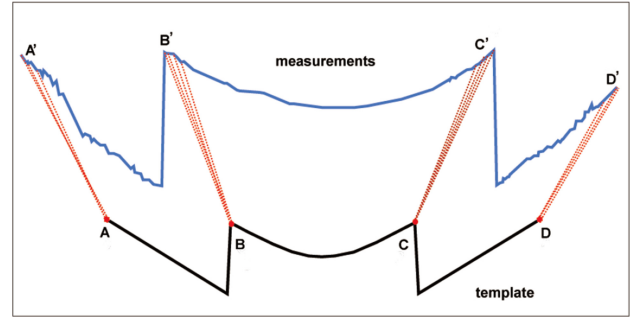


Fig. 8. Illustration of featured zone detection using method 1.

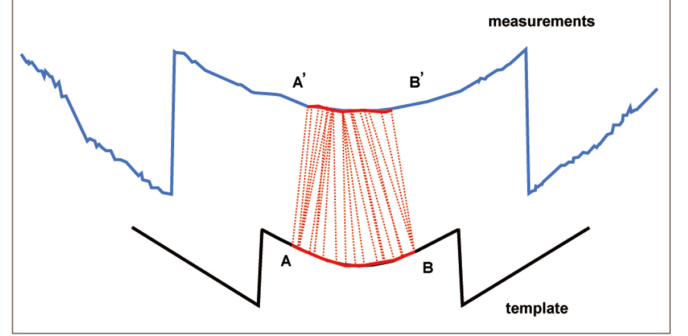


Fig. 9. Demonstration of method 2.

C. Bottom Detection

After $\cup$ zone detection, we need to identify the bottom of the $\cup$ zone which determines the tag orders. Accurate bottom detection is nontrivial because real measurements are usually noisy. In this article, we propose four methods to handle different situations and finally design a hybrid method that combines the advantages of these four methods.

Method 1: The first method is based on the symmetricity of the template. As shown in Fig. 8, we use four points (A, B, C, and D) in the template as reference points. Using subsequence DTW, we can find four small sets which correspond to these four reference points. We first calculate the average timestamp of each set. Then, the bottom point is decided to be the average timestamps of these four sets. This process can be described as follows:

$$T_1 = \frac{1}{4} \sum_{i=1}^4 \left(\frac{1}{N_i} \sum_{k=1}^{N_i} t_{i,k} \right) \quad (8)$$

where $t_{i,k}$ is the time stamp of k th data point in the i th set and N_i is the size of that set.

Method 2: Method 1 only works well when the measurements align well with the template, which is usually not the case in reality. In method 2, we choose a region to estimate the timestamp of the bottom. As shown in Fig. 9, we choose a region starting from A to B, then we find their corresponding region (A' to B') in the measurements. The timestamp of the bottom equals to the average timestamps of the points in A'-B'. The timestamp of bottom in method 2 can be expressed by

$$T_2 = \frac{\sum_{i=1}^N t_i}{N} \quad (9)$$

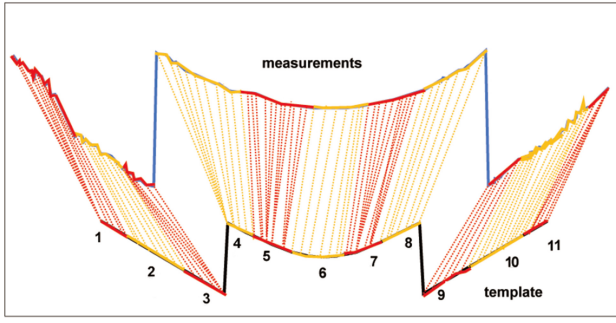


Fig. 10. Demonstration of method 3.

where N is the number of samples in the region and t_i is the timestamp of the i th sample.

Method 3: The former two methods are based on the symmetry of the $\cup$ region. However, if the moving speed of the reader varies, the $\cup$ region is not symmetric. To overcome this problem, we propose method 3. In order to mitigate the effect of changing speed, we add different weights to different samples. First, we divide the template into m consecutive regions with equal length and let R_i denote the i th region. In *TagSort*, m is empirically set to 11. Then, we assume the corresponding region of R_i in the measurements is R'_i which contains N_i samples. The timestamps of each sample in R'_i are $t_{i,1}, t_{i,2}, \dots, t_{i,N_i}$. Finally, the timestamp of bottom in method 3 can be calculated as follows:

$$\begin{cases} T_+ = \frac{1}{\sum_{i=1}^{11} N_i} \sum_{i=1}^{11} N_i \bar{T}_i \\ T_- = \frac{1}{\sum_{i=1}^{11} \frac{1}{N_i}} \sum_{i=1}^{11} \frac{1}{N_i} \bar{T}_i \\ T_3 = \frac{T_+ + T_-}{2} \end{cases} \quad (10)$$

where $\bar{T}_i$ represents the average timestamp in region R'_i . Fig. 10 illustrates the process of method 3.

The aforementioned three methods apply to different scenarios. The first method works well when the moving speed of RFID reader is steady. The second method works well when sufficient samples around the bottom point are detected and the reader is moving at a constant speed. The third method works well when the moving speed of RFID reader is varying. However, the real measurements are always not perfect and hard to estimate. To this end, we propose a hybrid method that combines these three methods together. The idea is simple but effective by adding different weights to different methods.

The weights of T_1 and T_2 are defined as follows:

$$w_1 = \exp\left(-\frac{1}{\sum_{k=1}^4 N_k} \sum_{i=1}^4 \sum_{k=1}^{N_i} (2\pi - \theta_{i,k})\right), \quad w_1 \in (0, 1] \quad (11)$$

where $\theta_{i,k}$ is the phase of the sample at $t_{i,k}$ time. If all reference points are close to 2π , which means that the phase profiles collected match the theoretical template very well, and the w_1 will become close to one. Therefore, the weight w_1 denotes the goodness of matching between the theoretical template and reference points (four reference points in method 1). When w_1 is closer to 1, the reference points match better with the

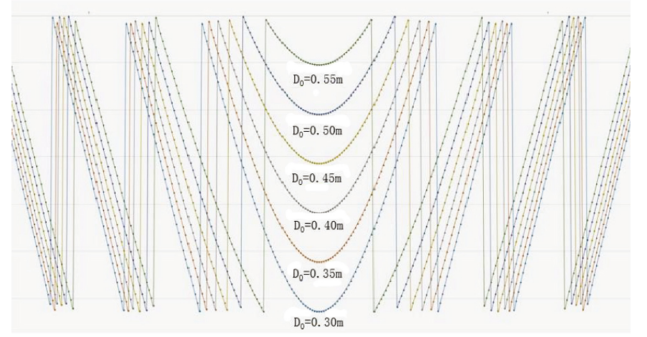
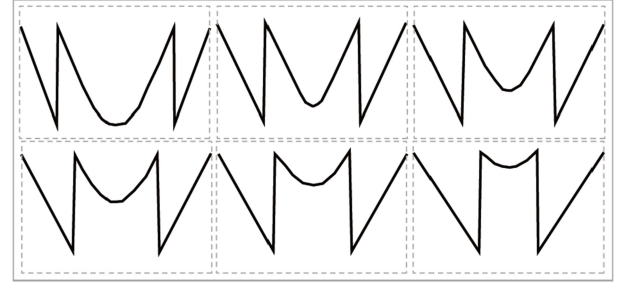
Fig. 11. Changes of $\cup$ zone.

Fig. 12. Final templates.

template. In this case, the higher reliability of the predicted value for both methods 1 and 2. Correspondingly, the weight of T_1 and T_2 will be higher.

The weight of T_3 is defined as follows:

$$w_2 = \exp\left(-\left|\frac{\sum_{k=1}^5 L_k}{\sum_{k=7}^{11} L_k} - 1\right|\right), \quad w_2 \in (0, 1]. \quad (12)$$

w_2 is defined in this way so that the closer w_2 is 0, the more irregular the moving speed of the reader. Correspondingly, we should attribute more weight to T_3 because methods 1 and 2 do not work well in this case. Therefore by combining all methods, the final hybrid bottom point calculation is defined as follows:

$$T_4 = \frac{w_1 w_2 T_1 + w_1 w_2 T_2 + w_1 (1 - w_2) T_3}{2 w_1 w_2 + w_1 (1 - w_2)}. \quad (13)$$

Once the bottom time of every tag is determined, the order of all tags can be easily obtained by ordering the timestamps from the earliest to the latest.

D. Consideration on Robustness

1) Dynamic Template Selection: In practice, the distance between the tags and the RFID reader can change, which affects the template generated and can make our detection unreliable. Fig. 11 shows that when D_0 changes, the template will change, especially the height of the $\cup$ zone. We also observe that the shape changes of the featured zone caused by the change of D_0 is quasi-periodic with a quasi-period of about 0.3 m.

To overcome this issue, we use multiple templates during $\cup$ zone detection. Specifically, we will run subsequent DTW on each of the template and then choose the best template that is the one producing the minimum distance. However, more

Algorithm 1 Optimal Template Searching

Input: RFID phase profile X and a set of template sequences $Y_0, Y_1, \dots, Y_K$

Output: the optimal template Y_{opt}

```

1: for  $i$  in  $\{0, 1, 2, \dots, K\}$  do
2:    $Subs = SDTW(X, Y_i)$ 
3:    $cost[i] = DTW(Subs[0], Y_i)$ 
4: end for
5:  $Y_{opt} = Y_0$ 
6:  $leastCost = cost[0]$ 
7: for  $i$  in  $\{0, 1, 2, \dots, K\}$  do
8:   if  $cost[i] < leastCost$  then
9:      $leastCost = cost[i]$ 
10:     $Y_{opt} = Y_i$ 
11:   end if
12: end for
13: return  $Y_{opt}$ 

```

templates mean longer computation time. In practice, we find that the six templates shown in Fig. 12 are enough to achieve high accuracy.

Algorithm 1 below shows how to pick up the optimal template. Notice that $SDTW(X, Y_i)$ refers to the process of subsequence DTW mentioned in Section IV-B.

2) *Hybrid Sorting Process:* We find that when two tags are close enough, the bottom of each tag is difficult to localize accurately because of the signal interference between tags. In particular, we find that the system does not work well when the timestamp difference of two tags is smaller than 200 ms. Meanwhile, we also observe that when two tags are close to each other, their phase profiles are similar because their surrounding environments are similar. In this case, we use the Pearson correlation coefficient to determine their orders. In statistics, the Pearson correlation coefficient is a statistic that describes the linear correlation between two random variables, whose value is between $[-1, +1]$. The definition is as follows:

$$\text{corr}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (14)$$

where x_i and y_i are two sequences, and $\bar{y}$ and $\bar{x}$ are the average values of each sequence. The principle of the proposed strategy is described as follows. We use the $\cup$ zone of X as the template. Then, we aim to find the region in Y that gives the highest correlation coefficient from the two sides of $\cup$ zone in Y . Finally, if the maximum correlation calculated from the left side is greater than the right side maximum correlation, then X is in front of Y , and vice versa. If they are equal, then X and Y are parallel. The details of the proposed strategy are explained in Algorithm 2.

Finally, the hybrid sorting process contains two stages. In the first stage, the $\cup$ zone and bottom detection using 13 will be performed to find the rough order among all tags. For those which are too close and unreliable using only the bottom detection, the correlation-based sorting using Algorithm 2 will be used to further improve the sorting accuracy.

Algorithm 2 Determine the Order of Two Close Tags

Input: Sequence X and its $\cup$ zone $X(a, b)$, Sequence Y , step length s

Output: the order of two tags

```

1: find  $Y(c)$  and  $Y(d)$  in  $Y$  that have the closet timestamps with  $X(a)$  and  $X(b)$ 
   find  $Y(e)$  in  $Y$  that is 200ms later than  $Y(c)$ 
   find  $Y(f)$  in  $Y$  that is 200ms earlier than  $Y(d)$ 
    $i = 0, j = c, k = d;$ 
2: repeat
3:    $R1(i) = \text{corr}(X(a, b), Y(j, k))$ 
4:    $j = j - s$ 
5:    $k = k - s$ 
6:    $i = i + 1$ 
7: until  $j < e$ 
8:  $i = 0, j = c, k = d;$ 
9: repeat
10:   $R2(i) = \text{corr}(X(a, b), Y(j, k))$ 
11:   $j = j + s$ 
12:   $k = k + s$ 
13:   $i = i + 1$ 
14: until  $k > f$ 
15: if  $\max(R1) > \max(R2)$  then
16:   return -1 ( $X$  is in front of  $Y$ )
17: else if  $\max(R1) < \max(R2)$  then
18:   return 1 ( $Y$  is in front of  $X$ )
19: end if
20: return 0 ( $X$  and  $Y$  are parallel)

```

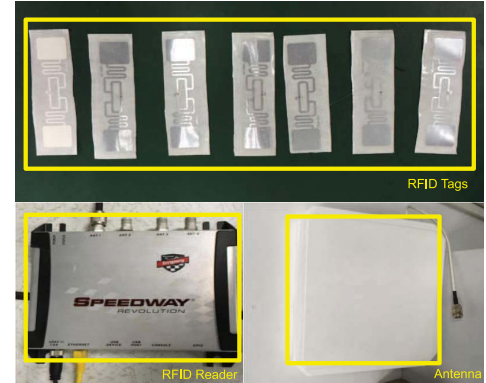


Fig. 13. Experimental settings.

V. EVALUATION

In this section, we provide a comprehensive evaluation of RFID-based indoor relative localization. We begin with a brief description of the experimental settings and then summarize the detailed results, including the different number of tags, different tag distances, and different vertical distances.

A. Experimental Settings

We implement the prototype of the RFID-based indoor relative localization using the commercially off-the-shelf devices. As shown in Fig. 13, the system prototype includes a commercial RFID reader *Impinj R420* and off-the-shelf *UHF* passive

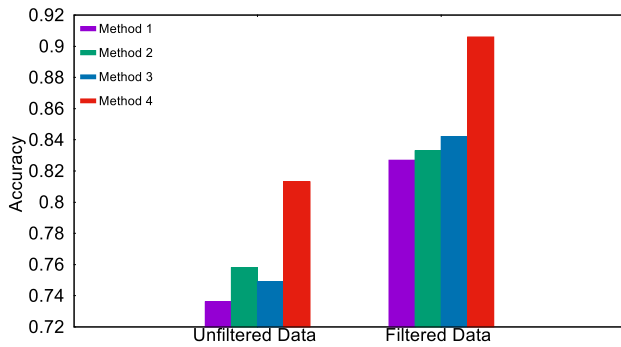


Fig. 14. Sorting accuracy on four different peak time estimation methods.

tag *Alien az-9662*, where the directional antenna of *Impinj* reader is *Laird A9028PCL*, with frequency range 865–956 MHz, and the gain 9 dBi.

B. Evaluation Results

1) *Accuracy on Four Different Peak Time Estimation Methods*: After the *subsequence DTW* mentioned in Section IV-B, the featured zone can be detected. The determination of the featured zone lays the foundation for the subsequent sorting process. Therefore, we can use the featured zone to find the bottom time and perform a preliminary tag sorting. We used four different peak time estimation methods to roughly sort the tags, and the accuracy of the classification is shown in Fig. 14. We can see that method 4 (hybrid scheme) effectively integrates the advantages of the other three methods and improves the overall accuracy of the sorting.

2) *Accuracy on Different Number of Tags and Different Tag Distances*: Considering the existence of signal interference between tags, in order to explore the effect of this mutual interference on the relative localization results, we performed a large number of sorting experiments under different number of tags and different tag distances. In the experiment, only when the order of the tags output by the algorithm is the same as the original order of the tags, can the phase localization be determined successfully. Conversely, as long as there is an incorrect tag order, the localization is considered to have failed.

Fig. 15 shows the effect of number of tags on the sorting success rate when tag distances are fixed. The solid line in the figure is the result of complete data processing, including the dual window moving filter preprocessing, feature zone detection based on DTW algorithm, preliminary sorting based on the peak time estimation, and fine-grained sorting based on the correlation coefficient. In contrast, the brown dashed lines in the figure differ only in that they are not preprocessed by dual window moving filter. It can be seen from the figure that the success rate of the sorting algorithm proposed in this article is generally between 90% and 100%. It can be seen that when the tag distance is relatively close, the number of tags has little influence on the sorting success rate. When the number is larger, the sorting success rate will gradually decrease as the number of tags increases, but it is not obvious overall. Comparing the trend of the solid line and the dotted line, it can be seen that when the tag distance is large, the filtering

work becomes more important. The sorting success rate and stability obviously decline without the preprocessing of filtering, while the sorting success rate after filtering can remain stable. Overall, the dual window moving average filter preprocessing can increase the sorting success rate by more than 15%. Comparing preliminary tag sorting based on peak time estimation (the sorting results are shown in Fig. 14), the fine-grained sorting based on the correlation coefficient method increases the overall ranking success rate by about 8%.

Fig. 16 shows the effect of the tag distance on the sorting success rate when the number of tags has a fixed value. It can be seen that the tag distance has little effect on the sorting success rate, indicating that the fine-grained sorting process based on the correlation coefficient works well, and as the tag distance increases, it has less influence on the sorting success rate.

3) *Accuracy on Different Vertical Distances*: In order to make our solution more robust and flexible in more complex situations, where the vertical distance D_0 in Fig. 1 between tags and reader are not constant, we applied a *vertical distance adaptive strategy*. We changed the value of the vertical distance D_0 and repeated all experiments in $D_0 = 10$ cm, $D_0 = 20$ cm, $D_0 = 30$ cm, $D_0 = 60$ cm, $D_0 = 80$ cm, and $D_0 = 120$ cm, respectively. The experimental results after applying the *vertical distance adaptive strategy* are shown in Fig. 17.

It can be seen that after applying the strategy, the sort success rate is significantly improved under the different vertical distances, indicating that the strategy is indeed working. However, we noticed that after the vertical distance reaches 80 cm, even if the strategy is applied, the order success rate is still not higher than 80%. We observe the waveform diagram of the corresponding phase time series and find that when the vertical distance is too long, the reader is always between the two adjacent equal-phase wavefronts of the tag echo, so the stable feature segment cannot be formed. Therefore, the subsequence matching algorithm will be invalidated, which is a point that the current algorithm cannot solve. This requires that the actual vertical distance between the reader and the tag should not be too far, otherwise the stable sequencing success rate cannot be guaranteed.

4) *Comparison With Related Works*: In this article, subsequence DTW, dynamic template selection, and fine-grained correlation-based ordering are applied, which greatly improves the localization accuracy and robustness of the algorithm. We compared the results with the state-of-the-art works using RSSI. OTrack [7] is a typical representative of the relative localization system using the RSSI signal. It can be seen from Figs. 18 and 19 that when the number of tags is only 2, the result of the relative localization using RSSI is comparable with *TagSort*. However, as the number of tags increases, the accuracy drops sharply due to multipath effect, signal instability, and environmental interference, but the *TagSort* accuracy drops only slightly. As the distance between tags increases, using RSSI the accuracy improves a little and *TagSort* remains high accuracy and is more robust. On average, the RSSI method achieves 47.6% accuracy, while *TagSort* achieves averages accuracy of over 95%.

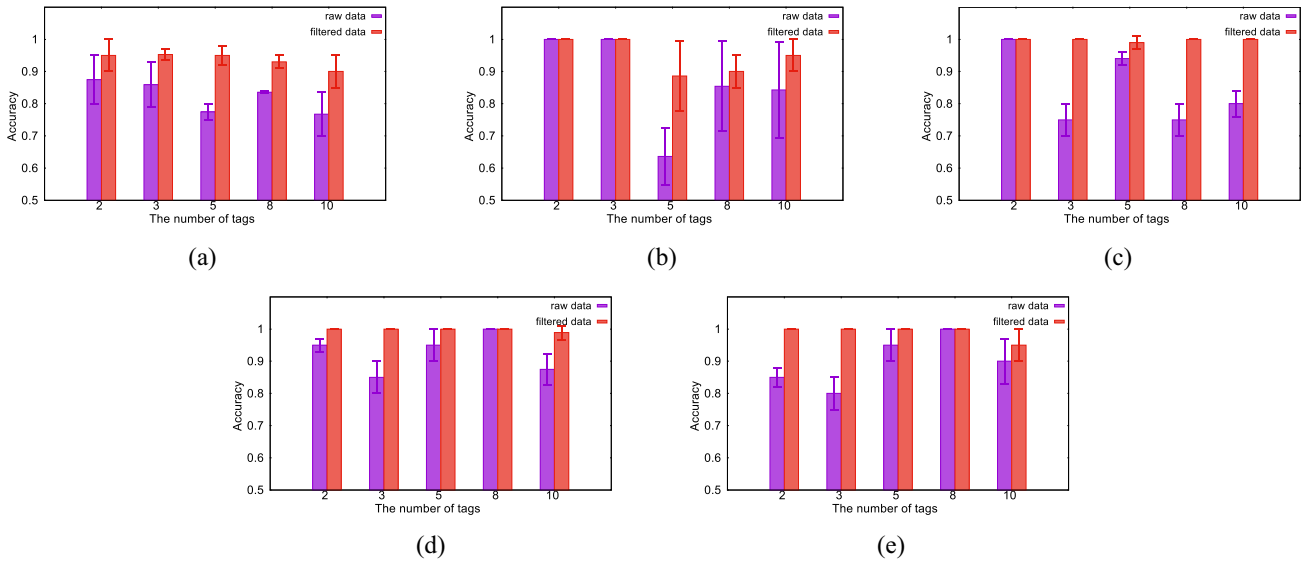


Fig. 15. Effect of the number of tags on sorting. Tags distances (a) $D = 2$ cm, (b) $D = 5$ cm, (c) $D = 8$ cm, (d) $D = 10$ cm, and (e) $D = 15$ cm.

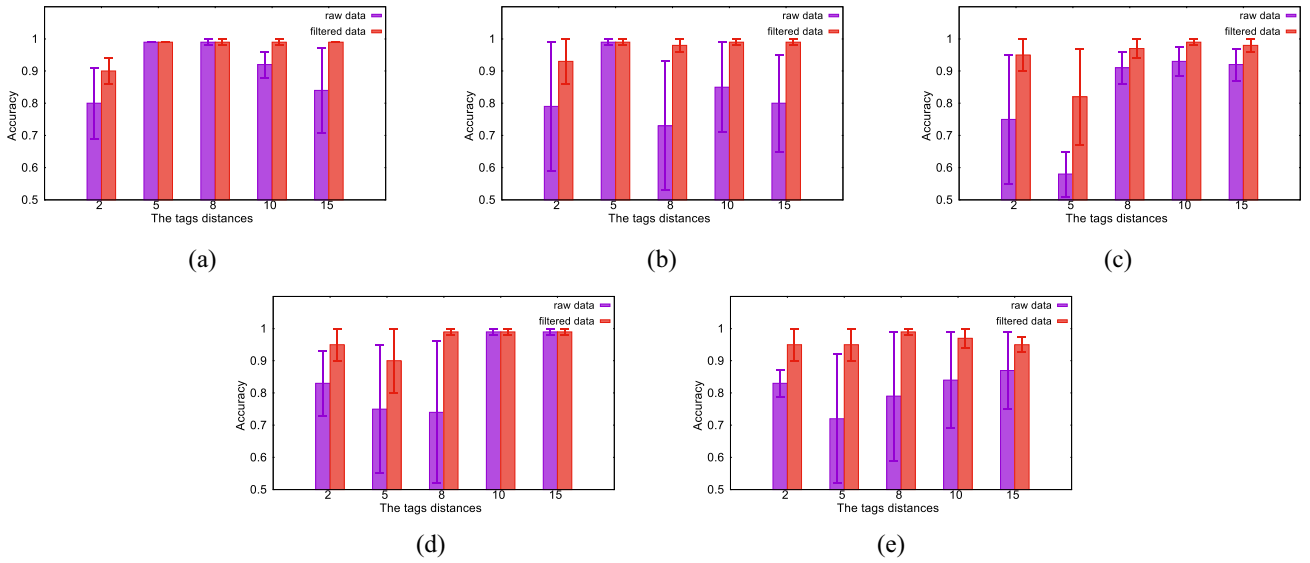


Fig. 16. Effect of the tags distances on sorting. Number of tags (a) $N = 2$, (b) $N = 3$, (c) $N = 5$, (d) $N = 8$, and (e) $N = 10$.

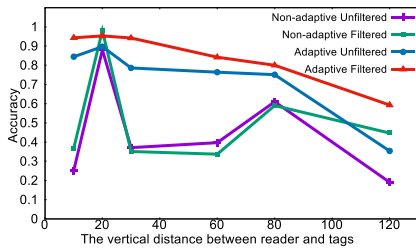


Fig. 17. Vertical distance adaptive strategy.

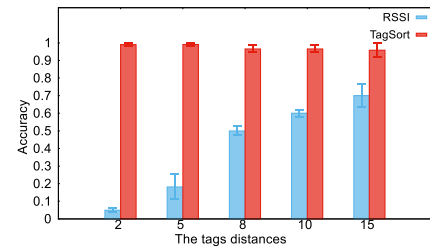


Fig. 18. RSSI versus TagSort by varying distances.

VI. LIMITATIONS AND FUTURE WORK

The relative localization scheme proposed in this article does have strong robustness and high localization accuracy, and it has certain application requirements in indoor scenes where relative technologies are required, such as factory

production lines, the luggage turntable of the airport, the bookshelf of the library, and the shelves of the shopping mall. However, the following aspects need to be optimized and improved in future work.

- 1) The current relative localization only involves the sorting of single-row tags, without considering the

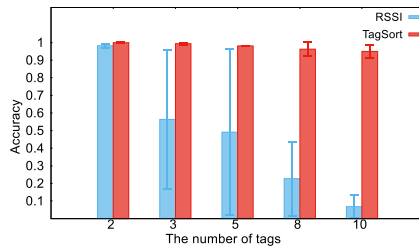


Fig. 19. RSSI versus TagSort by varying tags numbers.

relative localization of tags in multiple rows or even multidimensional directions, which limits the adaptability of the scheme in different application scenarios.

- 2) In this article, we use a single antenna for data acquisition, and the resulting data is not rich enough compared with multiple antennas or even multiple tags. If we can collect signal data from antennas in different positions and directions, it will be of great benefit to improve the accuracy and stability of the localization scheme.
- 3) When the vertical distance is large, the reader is always between two adjacent equal-phase wavefronts of the tag echo, and the tag phase time sequence acquired by the reader cannot form the expected feature zone, so that the follow-up algorithm fails. In future work, when the vertical distance is large, we use other strategies to solve this problem, which will improve the sorting stability and reduce the dependence on the data acquisition methods.

VII. CONCLUSION

With the continuous development of IoT technology, localization technology will be more and more applied in various aspects of modern society, making more and more important contributions to human society. Based on the RFID phase information, in this article, we apply the dynamic programming algorithm idea and the correlation coefficient method to design and implement a new scheme for relative localization of RFID tags. According to the difference of vertical distance between reader and tags, a vertical distance adaptive strategy based on the template selection was introduced. Experiments show that this new scheme achieved higher localization accuracy (95% sorting accuracy), better robustness, and better adaptability to environmental noise interference and scene changes. In addition, this solution is easy to implement, simple to operate, and easy to deploy to the work environment on a large scale.

REFERENCES

- [1] A. Al-Ali, F. A. Aloul, N. R. Aji, A. A. Al-Zarouni, and N. H. Fakhro, "Mobile RFID tracking system," in *Proc. 3rd Int. Conf. Inf. Commun. Technol. Theory Appl. (ICTTA)*, IEEE, Damascus, Syria, 2008, pp. 1–4.
- [2] H.-Y. Chien, "SASI: A new ultralightweight RFID authentication protocol providing strong authentication and strong integrity," *IEEE Trans. Dependable Secure Comput.*, vol. 4, no. 4, pp. 337–340, Oct.–Dec. 2007.

- [3] L. Yang, Y. Chen, X.-Y. Li, C. Xiao, M. Li, and Y. Liu, "Tagoram: Real-time tracking of mobile RFID tags to high precision using COTS devices," in *Proc. Int. Conf. Mobile Comput. Netw.*, Maui, HI, USA, 2014, pp. 237–248.
- [4] S. Jeschke, C. Brecher, H. Song, and D. B. Rawat, *Industrial Internet of Things: Cybermanufacturing Systems* (Springer Series in Wireless Technology). New York, NY, USA: Springer, 2017.
- [5] H. Song, R. Srinivasan, T. Sookoor, and S. Jeschke, *Smart cities: Foundations, Principles, and Applications*. Hoboken, NJ, USA: Wiley, 2017.
- [6] T. Liu, L. Yang, Q. Lin, Y. Guo, and Y. Liu, "Anchor-free backscatter positioning for RFID tags with high accuracy," in *Proc. IEEE INFOCOM*, Toronto, ON, Canada, 2014, pp. 379–387.
- [7] L. Shangguan, Z. Li, Z. Yang, M. Li, Y. Liu, and J. Han, "Otrack: Towards order tracking for tags in mobile RFID systems," *IEEE Trans. Parallel Distrib. Syst.*, vol. 25, no. 8, pp. 2114–2125, Aug. 2014.
- [8] L. Shangguan and K. Jamieson, "The design and implementation of a mobile RFID tag sorting robot," in *Proc. Int. Conf. Mobile Syst. Appl. Services*, Singapore, 2016, pp. 31–42.
- [9] P. V. Nikitin, R. Martinez, S. Ramamurthy, H. Leland, G. Spiess, and K. V. S. Rao, "Phase based spatial identification of UHF RFID tags," in *Proc. IEEE Int. Conf. Rfid.*, Orlando, FL, USA, 2010, pp. 102–109.
- [10] L. M. Ni, Y. Liu, Y. C. Lau, and A. P. Patil, "LANDMARC: Indoor location sensing using active RFID," *Wireless Netw.*, vol. 10, no. 6, pp. 701–710, 2004.
- [11] J. Wang and D. Katabi, "Dude, where's my card? RFID positioning that works with multipath and non-line of sight," in *Proc. ACM SIGCOMM Conf. SIGCOMM*, Hong Kong, 2013, pp. 51–62.
- [12] S. Azzouzi, M. Cremer, U. Dettmar, R. Kronberger, and T. Knie, "New measurement results for the localization of UHF RFID transponders using an angle of arrival (AoA) approach," in *Proc. IEEE Int. Conf. RFID.*, Orlando, FL, USA, 2011, pp. 91–97.
- [13] Y. Zhang, M. G. Amin, and S. Kaushik, "Localization and tracking of passive RFID tags based on direction estimation," *Int. J. Antennas Propag.*, vol. 2007, Art no. 17426, Oct. 2007.
- [14] L. Shangguan, Z. Yang, A. X. Liu, Z. Zhou, and Y. Liu, "Relative localization of RFID tags using spatial-temporal phase profiling," in *Proc. NSDI*, Oakland, CA, USA, 2015, pp. 251–263.
- [15] K. Kaemarungsi and P. Krishnamurthy, "Modeling of indoor positioning systems based on location fingerprinting," in *Proc. Joint Conf. IEEE Comput. Commun. Soc.*, vol. 2, Hong Kong, 2004, pp. 1012–1022.
- [16] M. Zhou, Y. Tang, Z. Tian, L. Xie, and W. Nie, "Robust neighborhood graphing for semi-supervised indoor localization with light-loaded location fingerprinting," *IEEE Internet Things J.*, vol. 5, no. 5, pp. 3378–3387, Oct. 2018.
- [17] S. Alletto *et al.*, "An indoor location-aware system for an IoT-based smart museum," *IEEE Internet Things J.*, vol. 3, no. 2, pp. 244–253, Apr. 2016.
- [18] D. Liang, Z. Zhang, and M. Peng, "Access point reselection and adaptive cluster splitting-based indoor localization in wireless local area networks," *IEEE Internet Things J.*, vol. 2, no. 6, pp. 573–585, Dec. 2015.
- [19] C. Luo, L. Cheng, M. C. Chan, Y. Gu, J. Li, and Z. Ming, "Pallas: Self-bootstrapping fine-grained passive indoor localization using WiFi monitors," *IEEE Trans. Mobile Comput.*, vol. 16, no. 2, pp. 466–481, Feb. 2017.
- [20] C. Luo, H. Hong, M. C. Chan, J. Li, X. Zhang, and Z. Ming, "MPiLoc: Self-calibrating multi-floor indoor localization exploiting participatory sensing," *IEEE Trans. Mobile Comput.*, vol. 17, no. 1, pp. 141–154, Jan. 2018.
- [21] J. Huiting, H. Flisijn, A. B. J. Kokkeler, and G. J. M. Smit, "Exploiting phase measurements of EPC Gen2 RFID tags," in *Proc. IEEE Int. Conf. Rfid-Tech. Appl.*, Johor Bahru, Malaysia, 2013, pp. 1–6.
- [22] P. Zhang, J. Gummeson, and D. Ganesan, "Blink: A high throughput link layer for backscatter communication," in *Proc. 10th Int. Conf. Mobile Syst. Appl. Service*, 2012, pp. 99–112.
- [23] *Dynamic Time Warping*. Heidelberg, Germany: Springer, 2007.
- [24] D. Berndt and J. Clithord, "Using dynamic time warping to find patterns in sequences," in *Proc. Work. Notes Knowl. Disc. Databases (WS)*, Seattle, WA, USA, 1994, pp. 359–370.
- [25] J. Barth *et al.*, "Subsequence dynamic time warping as a method for robust step segmentation using gyroscope signals of daily life activities," in *Proc. Eng. Med. Biol. Soc.*, Osaka, Japan, 2013, p. 6744.



Jinjiang Lai received the B.S. degree from the University of South China, Hengyang, China, in 2018. He is currently pursuing the master's degree with the School of Computer and Software Engineering, Shenzhen University, Shenzhen, China.

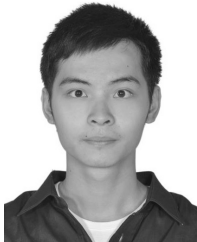
His current research interests include Internet of Things and machine learning.



Chengwen Luo (M'18) received the Ph.D. degree from the School of Computing, National University of Singapore, Singapore.

He is currently an Assistant Professor with the College of Computer Science and Software Engineering, Shenzhen University (SZU), Shenzhen, China. Before joining SZU, he was a Postdoctoral Researcher of CSE with the University of New South Wales, Sydney, NSW, Australia. He publishes regularly in top venues of networking, mobile computing, and Internet of Things, including ACM SenSys,

ACM/IEEE IPSN, the IEEE TRANSACTIONS ON MOBILE COMPUTING, the IEEE/ACM TRANSACTIONS ON NETWORKING, and ACM TOSN. His current research interests include mobile and pervasive computing, indoor localization, wireless sensor networks, and security aspects of Internet of Things.



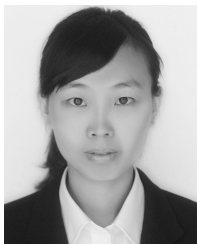
Jiawei Wu received the B.S. degree from South China Agricultural University, Guangzhou, China, in 2017. He is currently pursuing the master's degree with the School of Computer and Software Engineering, Shenzhen University, Shenzhen, China.

His current research interests include Internet of Things, data analysis, and deep learning.



Jianqiang Li received the B.S. and Ph.D. degrees from the South China University of Technology, Guangzhou, China, in 2003 and 2008, respectively.

He is a Professor with the College of Computer and Software Engineering, Shenzhen University, Shenzhen, China. He led three projects of the National Natural Science Foundation, and a project of the Natural Science Foundation of Guangdong Province, China. His current research interests include embedded systems and Internet of Things.



Jia Wang received the Ph.D. degree from the Department of Electronic Engineering, City University of Hong Kong, Hong Kong, in 2017.

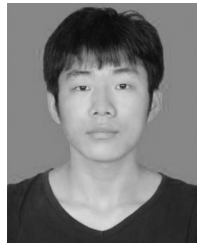
She is currently a Research Associate with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China. Her current research interests include security and privacy aspects of Internet of Things.



Jie Chen received the B.Sc. degree in electrical engineering from Taizhou University, Taizhou, China, in 2006, the M.Sc. degree in computer science from Zhejiang University, Hangzhou, China, in 2008, and the Ph.D. degree in computer science from the National University of Singapore (NUS), Singapore, in 2013.

He was a Postdoctoral Associate with Singapore-MIT Alliance for Research and Technology, Singapore, from 2013 to 2018. He is currently a Research-Track Associate Professor with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, China. His current research interests include machine learning, robotics, and multiagent system.

Prof. Chen was a recipient of the Dean's Graduate Research Excellence Award from NUS in 2013.



Gang Feng received the B.Sc. degree from Shenzhen University, Shenzhen, China, in 2015, and the master's degree from the College of Computer Science and Software Engineering, Shenzhen University, in 2017.

His current research interests include RFID-based localization, tracking, and various aspects of Internet of Things.



Houbing Song (S'05–M'12–SM'14) received the M.S. degree in civil engineering from the University of Texas at El Paso, El Paso, TX, USA, in December 2006, and the Ph.D. degree in electrical engineering from the University of Virginia, Charlottesville, VA, USA, in August 2012.

In August 2017, he joined the Department of Electrical, Computer, Software, and Systems Engineering, Embry-Riddle Aeronautical University, Daytona Beach, FL, USA, where he is currently an Assistant Professor and the Director of the Security and Optimization for Networked Globe Laboratory (www.SONGLab.us). He has served on the faculty of West Virginia University, Morgantown, WV, USA, from August 2012 to August 2017. In 2007, he was an Engineering Research Associate with Texas A&M Transportation Institute, Bryan, TX, USA. He has edited six books, including *Big Data Analytics for Cyber-Physical Systems: Machine Learning for the Internet of Things* (Elsevier, 2019), *Smart Cities: Foundations, Principles and Applications* (Hoboken, NJ, USA: Wiley, 2017), *Security and Privacy in Cyber-Physical Systems: Foundations, Principles and Applications* (Chichester, U.K.: Wiley-IEEE Press, 2017), *Cyber-Physical Systems: Foundations, Principles and Applications* (Boston, MA, USA: Academic Press, 2016), and *Industrial Internet of Things: Cybermanufacturing Systems* (Cham, Switzerland: Springer, 2016). He has authored over 100 articles. His current research interests include cyber-physical systems, cybersecurity and privacy, Internet of Things, edge computing, big data analytics, unmanned aircraft systems, connected vehicle, smart and connected health, and wireless communications and networking.

Dr. Song was a recipient of the Best Paper Award from the 2019 Integrated Communication, Navigation and Surveillance Technologies Conference, the prestigious Air Force Research Laboratory's Information Directorate (AFRL/RI) Visiting Faculty Research Fellowship, in 2018, and the Golden Bear Scholar Award, the highest campus-wide recognition for research excellence at West Virginia University Institute of Technology, in 2016. He serves as an Associate Technical Editor for the *IEEE Communications Magazine*. He is a Senior Member of ACM.